CBSE Class 12 Physics — Half-Yearly Predicted Paper (Set 1 of 10)

Section A: 1-Mark Questions (MCQ/Assertion-Reason)

Q1. Two point charges placed in a medium of dielectric constant 5 are at a distance r between them, experiencing an electrostatic force F. The electrostatic force between them in a vacuum at the exact same distance r will be: (a) $5F$ (b) F (c) $F/2$ (d) $F/5$ [PYQ 2021] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q2. The plates of a parallel plate capacitor are charged by a battery with V volts. After charging, the battery is disconnected and a dielectric slab with dielectric constant K is inserted between its plates. The potential difference across the plates of the capacitor will become: (a) Zero (b) $V/2$ (c) V/K (d) KV [PYQ 2021] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q3. The resistance of a metal wire fundamentally increases with increasing temperature on account of: (a) a massive decrease in free electron density. (b) a decrease in relaxation time. (c) an increase in mean free path. (d) an increase in the mass of the electron. [PYQ 2020] | **[Chapter 3: Current Electricity | Confidence: High]**

Q4. An electron is moving along the positive x-axis in a uniform magnetic field which is parallel to the positive y-axis. In what strict direction will the magnetic Lorentz force be acting on the electron? (a) Along $-x$ axis (b) Along $-z$ axis (c) Along $+z$ axis (d) Along $-y$ axis [PYQ 2023] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q5. A small compass needle of magnetic moment M and moment of inertia I is free to oscillate in a magnetic field B. It is slightly disturbed from its equilibrium position and then released. The time period of its oscillation is proportional to: (a) $\sqrt{M/B}$ (b) $\sqrt{MB/I}$ (c) $\sqrt{I/MB}$ (d) MB/I [PYQ 2023] | **[Chapter 5: Magnetism & Matter | Confidence: High]**

Q6. The magnetic flux linked with a coil (in Weber) is given by the equation $\phi = 5t^2 + 3t + 16$. The magnitude of the induced EMF in the coil at time $t = 4$ s will be: (a) 27 V (b) 43 V (c) 108 V (d) 210 V [PYQ 2021] | **[Chapter 6: Electromagnetic Induction | Confidence: Very High]**

Q7. If the frequency of an AC source is suddenly made 4 times its initial value, the inductive reactance of an inductor in the circuit will become: (a) 2 times (b) 3 times (c) 4 times (d) Unchanged [PYQ 2022] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q8. Which of the following electromagnetic waves has the highest frequency? (a) X-rays (b) Ultraviolet rays (c) Infrared rays (d) Gamma rays [PYQ 2022] | **[Chapter 8: Electromagnetic Waves | Confidence: Very High]**

Q9. A ray of monochromatic light of wavelength 600 nm propagates from air into a medium. If its wavelength in the medium becomes 400 nm, the refractive index of the medium is: (a) 1.4 (b) 1.5 (c) 1.6 (d) 1.8 [PYQ 2023] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Q10. In a Young's double slit experiment, the path difference at a certain point on the screen between two interfering waves is $1/8^{\text{th}}$ of the wavelength ($\lambda/8$). The ratio of the intensity at this point to that at the centre of a bright fringe is close to: (a) 0.80 (b) 0.74 (c) 0.94 (d) 0.85 [PYQ 2022] | **[Chapter 10: Wave Optics | Confidence: High]**

Q11. The work function for a given metal surface is $4.14\ \text{eV}$. The threshold wavelength for this metal surface is: (a) $4125\ \text{\AA}$ (b) $2062.5\ \text{\AA}$ (c) $3000\ \text{\AA}$ (d) $6000\ \text{\AA}$ [PYQ 2022] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Q12. The kinetic energy of a proton and that of an α-particle are $4\ \text{eV}$ and $1\ \text{eV}$, respectively. The ratio of the de-Broglie wavelengths associated with them will be: (a) $2:1$ (b) $1:1$ (c) $1:2$ (d) $4:1$ [PYQ 2020] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

For Questions 13 to 16, two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c), and (d) as given below. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false and R is also false.

Q13. Assertion (A): Work done in moving a charge around any closed path in an electrostatic field is always zero. **Reason (R):** Electrostatic force is a conservative force. [PYQ 2023] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q14. Assertion (A): Kirchhoff's junction rule is based on the law of conservation of charge. **Reason (R):** A resistor obeys Ohm's law while a diode does not. [PYQ 2022] | **[Chapter 3: Current Electricity | Confidence: High]**

Q15. Assertion (A): In Young's double slit experiment, all bright fringes are of exactly equal width. **Reason (R):** The fringe width depends directly upon the wavelength of light (λ) used, the distance of the screen from the plane of the slits (D), and inversely on the slit separation (d). [PYQ 2023] | **[Chapter 10: Wave Optics | Confidence: Very High]**

Q16. Assertion (A): The photoelectrons produced by a monochromatic light beam incident on a metal surface have a spread in their kinetic energies, ranging from zero to a maximum value. **Reason (R):** The energy of electrons emitted from deep inside the metal surface is lost in collisions with the other atoms in the metal before reaching the surface. [PYQ 2022] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Section B: 2-Mark Questions (Very Short Answer)

Q17. Depict the equipotential surfaces due to an isolated negative point charge ($-q$). Draw the corresponding electric field lines as well. [PYQ 2020] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q18. Define the term 'mobility' of charge carriers in a current-carrying conductor. How is the drift velocity of electrons in a conductor affected when the temperature of the conductor rises? [PYQ 2019, 2020] | **[Chapter 3: Current Electricity | Confidence: High]**

Q19. Define the terms 'magnetic susceptibility' and 'relative magnetic permeability'. Write the fundamental mathematical relation between them. [PYQ 2018] | **[Chapter 5: Magnetism & Matter | Confidence: Very High]**

Q20. How exactly is a displacement current produced between the plates of a parallel plate capacitor during the charging process? Write the mathematical expression for it. [PYQ 2020] | **[Chapter 8: Electromagnetic Waves | Confidence: Very High]**

Q21. An electron microscope fundamentally uses electrons accelerated by a high voltage of $50\ \mathrm{kV}$. Determine the de-Broglie wavelength inherently associated with these highly energetic electrons. [PYQ 2014] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: High]**

Section C: 3-Mark Questions (Short Answer)

Q22. Use Gauss's law to rigorously obtain an expression for the electric field due to an infinitely long thin straight wire possessing a uniform linear charge density λ. [PYQ 2023] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q23. A cell of emf E and internal resistance r is connected across a variable external resistor R. Plot a graph showing the variation of the terminal voltage V of the cell versus the current I drawn from it. Using this plot, show how the exact values of the emf of the cell and its internal resistance can be determined. [PYQ 2014] | **[Chapter 3: Current Electricity | Confidence: High]**

Q24. State the Biot-Savart law. Use this law to systematically derive the expression for the magnetic field on the axis of a current-carrying circular loop of radius R at a distance x from its centre. [PYQ 2016] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q25. State Lenz's law of electromagnetic induction. Explain with proper logical reasoning how Lenz's law is a direct consequence of the principle of conservation of energy. [PYQ 2014] | **[Chapter 6: Electromagnetic Induction | Confidence: Very High]**

Q26. An AC source generating a voltage $V = V_0 \sin(\omega t)$ is connected to a pure capacitor of capacitance C. Derive the expression for the current I flowing through it. Plot a graph showing the variation of capacitive reactance (X_C) with the change in frequency of the AC source. [PYQ 2014, 2015] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q27. Two monochromatic rays of light labelled '1' and '2' are incident normally on the face AB of an isosceles right-angled glass prism ABC. The refractive indices of the glass prism for rays '1' and '2' are 1.35 and 1.45 respectively. Trace the exact path of these rays after entering through the prism, providing mathematical justification. [PYQ 2014] | **[Chapter 9: Ray Optics | Confidence: High]**

Q28. State Huygens' principle of secondary wavelets. Using Huygens' construction, draw a diagram showing the passage of a plane wavefront from a denser medium into a rarer medium, and hence verify Snell's law of refraction. [PYQ 2015, 2022] | **[Chapter 10: Wave Optics | Confidence: Very High]**

Section D: 4-Mark Questions (Case-Study Based)

Q29. Case Study: Total Internal Reflection and Optical Fibres When a ray of light travels from an optically denser medium to a rarer medium, it bends away from the normal. As the angle of incidence increases, the angle of refraction also increases until the refracted ray grazes the interface. The angle of incidence for which this happens is called the critical angle (θ_c). If the angle of incidence is increased further, the ray is completely reflected back into the denser medium. This phenomenon is called Total Internal Reflection (TIR). Optical fibres use TIR to transmit light signals over long distances with negligible loss of energy. An optical fibre consists of a central core surrounded by a cladding. (A) What is the essential condition regarding the refractive indices of the core and the cladding for an optical fibre to function? (1 Mark) (B) Write the formula relating the critical angle (θ_c) to the refractive index (μ) of the denser medium with respect to the rarer medium. (1 Mark) (C) A point source of light is placed at the bottom of a tank filled with water of refractive index μ to a depth d. Calculate the area of the circular surface of water through which light from the source can emerge into the air above. (2 Marks) [Practice / PYQ 2022] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Q30. Case Study: Photoelectric Effect The photoelectric effect is the phenomenon of the emission of electrons from a metal surface when light of a sufficiently high frequency falls upon it. The minimum frequency required to cause emission is called the threshold frequency (ν_0). The maximum kinetic energy of the emitted photoelectrons depends on the frequency of the incident light and the nature of the material, but is completely independent of the intensity of the light. (A) How does the stopping potential theoretically change if the intensity of the incident light is doubled while keeping its frequency constant? (1 Mark) (B) Identify the specific physical constant that can be determined from the slope of a graph plotted between stopping potential (V_0) and the frequency (ν) of incident light. (1 Mark) (C) The work function of a metal is $2.31\ \mathrm{eV}$. Photoelectric emission occurs when light of frequency $6.4 \times 10^{14}\ \mathrm{Hz}$ is incident on the metal surface. Calculate the maximum kinetic energy (in eV) of the emitted electron. (Take $h = 6.63 \times 10^{-34}\ \mathrm{J\ s}$, $1\ \mathrm{eV} = 1.6 \times 10^{-19}\ \mathrm{J}$). (2 Marks) [PYQ 2017, 2022] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Section E: 5-Mark Questions (Long Answer)

Q31. (A) Derive the mathematical expression for the capacitance of a parallel plate capacitor having plate area A and plate separation d, with air present between the two plates. (B) A parallel plate capacitor of capacitance C is charged to a potential V volts by a battery. After some time, the battery is disconnected. A solid slab of dielectric constant K ($K > 1$) is now slowly introduced to completely fill the space between the plates. How will the following physical quantities be affected? (i) The capacitance of the capacitor. (ii) The electric field between the plates. (iii) The electrostatic energy stored in the capacitor. Justify your answers by writing the necessary mathematical expressions. [PYQ 2018, 2019] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q32. (A) Derive the expression for the magnetic force per unit length acting between two infinitely long straight parallel conductors carrying steady currents I_1 and I_2 separated by a perpendicular distance d. (B) Based on this derived expression, precisely define the SI unit of current (one Ampere). (C) A square current-carrying loop is placed near an infinitely long straight wire carrying a steady current I. The wire and the loop lie in the exact same plane, with one side of the square parallel to the wire. Will the net magnetic force exerted by the straight wire on the loop be attractive or repulsive if the current in the closest side of the loop flows in the same direction as the current in the straight wire? Justify briefly. [PYQ 2019, 2022] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q33. (A) Draw a neat and labelled ray diagram showing the refraction of a monochromatic ray of light through a triangular glass prism. (B) Deduce the standard expression for the refractive index μ of the material of the prism in terms of the angle of the prism A and the angle of minimum deviation δ_m. (C) Calculate the angle of emergence (e) of a ray of light incident normally on the face AB of an equilateral glass prism ABC of refractive index $\mu = \sqrt{3}$. Will the ray suffer total internal reflection at face AC? [PYQ 2020] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Answer Key

Q1: (a) $5F$. The force in a vacuum is K times the force in a dielectric medium. $F_{vac} = K \times F_{med} = 5F$. **Q2:** (c) V/K. Since the battery is disconnected, charge Q remains constant. New capacitance $C' = KC$. New potential $V' = Q/C' = Q/KC = V/K$. **Q3:** (b) a decrease in relaxation time. As temperature increases, the thermal speed and collision frequency of lattice ions increase, decreasing the average relaxation time between electron collisions, thereby increasing resistance. **Q4:** (b) Along $-z$ axis. Velocity $\vec{v}$ is along $+\hat{i}$, Magnetic field $\vec{B}$ is along $+\hat{j}$. Lorentz force $\vec{F} = q(\vec{v} \times \vec{B}) = (-e)(\hat{i} \times \hat{j}) = (-e)(\hat{k}) = -e\hat{k}$ (negative z-direction). **Q5:** (c) $\sqrt{I/MB}$. The time period of a magnetic dipole in a magnetic field is strictly given by $T = 2\pi\sqrt{I/MB}$. **Q6:** (b) $43\ \mathrm{V}$. Induced emf $|e| = d\phi/dt = \frac{d}{dt}(5t^2 + 3t + 16) = 10t + 3$. At $t = 4$ s, $|e| = 10(4) + 3 = 43\ \mathrm{V}$. **Q7:** (c) 4 times. Inductive reactance $X_L = 2\pi fL$. Since $X_L \propto f$, if frequency is 4 times, X_L becomes 4 times. **Q8:** (d) Gamma rays. They lie at the extreme short-wavelength, high-frequency end of the electromagnetic spectrum. **Q9:** (b) 1.5. Refractive index $\mu = \lambda_{vac}/\lambda_{med} = 600\ \mathrm{nm}/400\ \mathrm{nm} = 1.5$. **Q10:** (d) 0.85. Path difference $\Delta x = \lambda/8 \Rightarrow$ Phase difference $\Delta\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{8} = \pi/4$. Intensity $I = I_0 \cos^2(\Delta\phi/2) = I_0 \cos^2(\pi/8) \approx 0.85 I_0$. **Q11:** (c) $3000\ \backslash\mathrm{AA}$. $\lambda_0 = hc/W = 1242\ \mathrm{eV\ nm}/4.14\ \mathrm{eV} = 300\ \mathrm{nm} = 3000\ \backslash\mathrm{AA}$. **Q12:** (b) $1:1$.

$\lambda = h/\sqrt{2mK}$. Ratio
$\lambda_p/\lambda_\alpha = \sqrt{m_\alpha K_\alpha / m_p K_p} = \sqrt{(4m_p \times 1\text{ eV})/(m_p \times 4\text{ eV})} = \sqrt{4/4} = 1 : 1$.

Q13: (a) Both A and R are true and R is the correct explanation of A. **Q14:** (b) Both A and R are true but R is NOT the correct explanation of A. **Q15:** (a) Both A and R are true and R is the correct explanation of A. **Q16:** (a) Both A and R are true and R is the correct explanation of A.

Q17: Concentric spherical shells spaced progressively further apart outwards. Electric field lines are straight radial lines pointing strictly *inwards* towards the negative charge $-q$. **Q18:** Mobility (μ) is the magnitude of drift velocity per unit applied electric field ($\mu = v_d/E$). When temperature rises, the relaxation time τ decreases due to more frequent collisions, which directly heavily decreases the drift velocity ($v_d = eE\tau/m$). **Q19:** Susceptibility (χ_m) measures how easily a substance gets magnetized in a magnetizing field. Relative permeability (μ_r) is the ratio of permeability of the material to free space. Relation: $\mu_r = 1 + \chi_m$. **Q20:** During charging, the changing electric field (dE/dt) between the plates causes a varying electric flux ($d\phi_E/dt$). This changing flux acts precisely like a current. $I_d = \epsilon_0(d\phi_E/dt)$. **Q21:** $\lambda = \frac{1.227}{\sqrt{V}}\text{ nm} = \frac{1.227}{\sqrt{50000}}\text{ nm} = \frac{1.227}{223.6}\text{ nm} \approx 0.0055\text{ nm} = 0.055 \text{\AA}$.

Q22: Consider a cylindrical Gaussian surface of radius r and length l around the wire. $\oint \vec{E} \cdot d\vec{S} = q_{enc}/\epsilon_0$. $E \times (2\pi rl) = (\lambda l)/\epsilon_0 \Rightarrow E = \frac{\lambda}{2\pi\epsilon_0 r}$. **Q23:** The plot of V vs I is a straight line with a negative slope, satisfying $V = E - Ir$. The y-intercept (when $I = 0$) directly gives the emf E. The absolute value of the slope gives the internal resistance r. **Q24:** Draw diagram showing loop, current I, radius R, axis x. $dB = \frac{\mu_0 I dl \sin 90^\circ}{4\pi(R^2+x^2)}$. Resolve components; vertical components cancel, axial components $dB\sin\theta$ add up. Integrate to get $B = \frac{\mu_0 I R^2}{2(R^2+x^2)^{3/2}}$. **Q25:** The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produces it. To move a magnet against this opposing induced magnetic field requires mechanical work. This mechanical work entirely transforms into electrical energy (induced current), conserving energy. **Q26:** $V = V_0\sin(\omega t)$. $Q = CV \Rightarrow I = dQ/dt = d(CV_0\sin\omega t)/dt = \omega CV_0\cos(\omega t) = I_0\sin(\omega t + \pi/2)$. Thus, current leads by $\pi/2$. Graph of $X_C = 1/\omega C$ vs ν is a rectangular hyperbola strictly decreasing. **Q27:** Both rays hit the hypotenuse at angle $i = 45^\circ$. Critical angle for Ray 1: $\sin\theta_{c1} = 1/1.35 = 0.74 \Rightarrow \theta_{c1} > 45^\circ$ (Ray 1 totally refracts out). Ray 2: $\sin\theta_{c2} = 1/1.45 = 0.68 \Rightarrow \theta_{c2} < 45^\circ$ (Ray 2 undergoes TIR). **Q28:** Every point on a wavefront acts as a secondary source. Draw incident wavefront AB. Let speed in denser be v_1, rarer be v_2 ($v_2 > v_1$). Construct refracted wavefront CD. Prove $\sin i/\sin r = (BC/v_1)/(AD/v_2) = v_1/v_2 = \mu_2/\mu_1$.

Q29: (A) The refractive index of the core must be strictly *greater* than that of the cladding. (B) $\sin\theta_c = 1/\mu$. (C) Light forms a cone of semi-angle θ_c. Radius of surface $r = d\tan\theta_c$. From $\sin\theta_c = 1/\mu$, $\tan\theta_c = 1/\sqrt{\mu^2 - 1}$. Area $A = \pi r^2 = \frac{\pi d^2}{\mu^2-1}$. **Q30:** (A) The stopping potential remains entirely unchanged because it depends on frequency, not intensity. (B) Planck's constant divided by electron charge (h/e). (C) $E = h\nu = \frac{6.63\times10^{-34}\times6.4\times10^{14}}{1.6\times10^{-19}}\text{ eV} \approx 2.65\text{ eV}$. $K_{max} = E - W = 2.65 - 2.31 = 0.34\text{ eV}$.

Q31: (A) $E = \sigma/\epsilon_0 = Q/A\epsilon_0$. $V = Ed = Qd/A\epsilon_0$. Capacitance $C = Q/V = \epsilon_0 A/d$. (B) (i) Capacitance increases K times ($C' = KC$). (ii) Since charge Q is constant, $E' = V'/d = (V/K)/d = E/K$. Electric field decreases by factor K. (iii) Energy $U' = Q^2/2C' = Q^2/2(KC) = U/K$. Energy strictly decreases by factor K. **Q32:** (A) Magnetic field due to Wire 1 at Wire 2 is $B_1 = \mu_0 I_1/2\pi d$. Force on length L of Wire 2 is $F = I_2 L B_1 \sin 90^\circ = \mu_0 I_1 I_2 L/2\pi d$. Force per unit length $f = \mu_0 I_1 I_2/2\pi d$. (B) One Ampere is that steady current which, flowing in two infinitely long straight parallel wires $1\ \mathrm{m}$ apart in vacuum, produces a force of $2 \times 10^{-7}\ \mathrm{N}$ per metre of length. (C) Attractive. The attractive force on the closer parallel arm (carrying current in the same direction) is far stronger than the repulsive force on the farther parallel arm (current in opposite direction). **Q33:** (A) Proper diagram showing incident ray, refracted ray inside prism, emergent ray, angle of deviation δ, and angles i, e, r_1, r_2. (B) At minimum deviation δ_m, $i = e$ and $r_1 = r_2 = r$. Since $A = r_1 + r_2 \Rightarrow r = A/2$. And $\delta_m = i + e - A \Rightarrow i = (A + \delta_m)/2$. Snell's law: $\mu = \sin i/\sin r = \sin((A + \delta_m)/2)/\sin(A/2)$. (C) Ray hits face AB normally ($i = 0$), travels undeviated ($r_1 = 0$). Hits face AC at 60° (since $A = 60^\circ$). For total internal reflection, $\sin\theta_c = 1/\mu = 1/\sqrt{3} \approx 0.577$. Angle of incidence at AC is 60°. $\sin 60^\circ = \sqrt{3}/2 \approx 0.866$. Since $0.866 > 0.577$ ($i > \theta_c$), the ray suffers Total Internal Reflection at face AC. Angle of emergence from face BC would require further geometric tracing, but it does not emerge from AC.

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